A DESCENT DATUM GIVES A COSIMPLICIAL MODULE, FUNCTORIALLY

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Allowed standard prerequisites (editorial).

- Cosimplicial objects and the list of cosimplicial relations. Simplicial, Section simplicial-section-cosimplicial-object and Remark simplicial-remark-relations-cosimplicial, invoked at proof lines 626-627 and 660-662. Supplies the axioms that must be verified.
- The cocycle condition for a descent datum. Same chapter, the description of descent data for a ring map, used at proof lines 632-654. Supplies the identity that makes the second codegeneracy-coface relation hold.
- The tensor constructions N_{n,n} attached to a ring map. Same chapter, the notation for the tensor powers of A over R acting on N, used in the statement. Supplies the objects in each cosimplicial degree.

Source reference. Original remark 016L: sources/stacks-chow/tags.txt, source lines 1548–1548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 016I: sources/stacks-chow/tags.txt, source lines 1545–1545. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $R \to A$ be a ring map. Given a descent datum (N, φ) we can associate to it a cosimplicial $(A/R)_\bullet$-module $N_\bullet$[1] by the rules $N_n = N_{n,n}$ and given $\beta : [n] \to [m]$ setting we define*

$$N_\bullet(\beta) = (\varphi^m_{\beta(n)m}) \circ N_{\beta,n} : N_{n,n} \longrightarrow N_{m,m}.$$

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[1]We should really write $(N, \varphi)_\bullet$.

This procedure is functorial in the descent datum.

Proof. Here are the first few maps where $\varphi(n \otimes 1) = \sum \alpha_i \otimes x_i$

$$\begin{array}{lcccccc}
\delta_0^1 & : & N & \to & A \otimes N & n & \mapsto & 1 \otimes n \\
\delta_1^1 & : & N & \to & A \otimes N & n & \mapsto & \sum \alpha_i \otimes x_i \\
\sigma_0^0 & : & A \otimes N & \to & N & a_0 \otimes n & \mapsto & a_0 n \\
\delta_0^2 & : & A \otimes N & \to & A \otimes A \otimes N & a_0 \otimes n & \mapsto & 1 \otimes a_0 \otimes n \\
\delta_1^2 & : & A \otimes N & \to & A \otimes A \otimes N & a_0 \otimes n & \mapsto & a_0 \otimes 1 \otimes n \\
\delta_2^2 & : & A \otimes N & \to & A \otimes A \otimes N & a_0 \otimes n & \mapsto & \sum a_0 \otimes \alpha_i \otimes x_i \\
\sigma_0^1 & : & A \otimes A \otimes N & \to & A \otimes N & a_0 \otimes a_1 \otimes n & \mapsto & a_0 a_1 \otimes n \\
\sigma_1^1 & : & A \otimes A \otimes N & \to & A \otimes N & a_0 \otimes a_1 \otimes n & \mapsto & a_0 \otimes a_1 n
\end{array}$$

with notation as in Simplicial, Section 016I. We first verify the two properties $\sigma_0^0 \circ \delta_0^1 = \mathrm{id}$ and $\sigma_0^0 \circ \delta_1^1 = \mathrm{id}$. The first one, $\sigma_0^0 \circ \delta_0^1 = \mathrm{id}$, is clear from the explicit description of the morphisms above. To prove the second relation we have to use the cocycle condition (because it does not hold for an arbitrary isomorphism $\varphi : N \otimes_R A \to A \otimes_R N$). Write $p = \sigma_0^0 \circ \delta_1^1 : N \to N$. By the description of the maps above we deduce that p is also equal to

$$p = \varphi \otimes \mathrm{id} : N = (N \otimes_R A) \otimes_{(A \otimes_R A)} A \longrightarrow (A \otimes_R N) \otimes_{(A \otimes_R A)} A = N$$

Since φ is an isomorphism we see that p is an isomorphism. Write $\varphi(n \otimes 1) = \sum \alpha_i \otimes x_i$ for certain $\alpha_i \in A$ and $x_i \in N$. Then $p(n) = \sum \alpha_i x_i$. Next, write $\varphi(x_i \otimes 1) = \sum \alpha_{ij} \otimes y_j$ for certain $\alpha_{ij} \in A$ and $y_j \in N$. Then the cocycle condition says that

$$\sum \alpha_i \otimes \alpha_{ij} \otimes y_j = \sum \alpha_i \otimes 1 \otimes x_i.$$

This means that $p(n) = \sum \alpha_i x_i = \sum \alpha_i \alpha_{ij} y_j = \sum \alpha_i p(x_i) = p(p(n))$. Thus p is a projector, and since it is an isomorphism it is the identity.

To prove fully that $N_\bullet$ is a cosimplicial module we have to check all 5 types of relations of Simplicial, Remark 016L. The relations on composing σ's are obvious. The relations on composing δ's come down to the cocycle condition for φ. In exactly the same way as above one checks the relations $\sigma_j \circ \delta_j = \sigma_j \circ \delta_{j+1} = \mathrm{id}$. Finally, the other relations on compositions of δ's and σ's hold for any φ whatsoever. $\square$